

Library Management System - Easy Code

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<!DOCTYPE html>
<html>
<head>
  <title>Library Management System</title>
  <style>
    body{
      font-family: Arial;
      margin:20px;
    }
    .book{
      border:1px solid black;
      padding:10px;
      margin:10px;
    }
    button{
      margin:5px;
    }
  </style>
</head>
<body>

<h2>Library Management System</h2>

<input type="text" id="search" placeholder="Search Book">
<button onclick="searchBook()">Search</button>

<h3>Add Book (Admin)</h3>
<input type="text" id="title" placeholder="Book Title">
<input type="text" id="author" placeholder="Author">
<input type="number" id="qty" placeholder="Quantity">
<button onclick="addBook()">Add Book</button>

<h3>Book List</h3>
<div id="bookList"></div>

<script>
let books = [
  {id:1, title:"Harry Potter", author:"J.K Rowling", qty:3, available:3},
  {id:2, title:"1984", author:"George Orwell", qty:2, available:2}
];

function displayBooks(data = books){
  let output = "";

  data.forEach(book => {
    output +=
      <div class="book">
        <h3>${book.title}</h3>
        <p>Author: ${book.author}</p>
        <p>Available: ${book.available}</p>

        <button onclick="reserveBook(${book.id})">Reserve</button>
        <button onclick="borrowBook(${book.id})">Borrow</button>
        <button onclick="returnBook(${book.id})">Return</button>
      </div>
    `;
  });

  document.getElementById("bookList").innerHTML = output;
}

function reserveBook(id){
  let book = books.find(b => b.id == id);

  if(book.available > 0){
    alert("Book Reserved");
  } else {
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        alert("Book Not Available");
    }
}

function borrowBook(id){
    let book = books.find(b => b.id == id);

    if(book.available > 0){
        book.available--;
        alert("Book Borrowed");
    } else {
        alert("No Books Available");
    }

    displayBooks();
}

function returnBook(id){
    let book = books.find(b => b.id == id);

    if(book.available < book.qty){
        book.available++;
        alert("Book Returned");
    }

    displayBooks();
}

function addBook(){
    let title = document.getElementById("title").value;
    let author = document.getElementById("author").value;
    let qty = parseInt(document.getElementById("qty").value);

    books.push({
        id: books.length + 1,
        title:title,
        author:author,
        qty:qty,
        available:qty
    });

    displayBooks();
}

function searchBook(){
    let text = document.getElementById("search").value.toLowerCase();

    let result = books.filter(book =>
        book.title.toLowerCase().includes(text)
    );

    displayBooks(result);
}

displayBooks();
</script>

</body>
</html>

```